

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 2

Total Marks:40

Annexure A

Integrated Experiments

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Student 24: Preethi R | Register Number: 192511352

INTEGRATED EXPERIMENT SCENARIO

Manufacturing - Equipment Failure Prediction

Context: A manufacturing plant wants an integrated pipeline that first predicts equipment failure from sensor records (vibration, temperature, operating hours, load) and then recommends a maintenance action using a learning agent.

Question 1 - Decision Tree vs Neural Network Classifier

Implement a Decision Tree classifier and a Neural Network (Multi-Layer Perceptron) classifier on the same equipment dataset. Compare their Accuracy, Precision, Recall and F1-Score. Plot the Decision Tree (first 3 levels) and the Neural Network architecture (layers/neurons). Justify which model you would deploy for real-time failure alerts and why.

Question 2 - Reinforcement Learning for Maintenance Scheduling

Using the failure-risk output from Experiment 1 as the initial state, implement a Q-Learning agent that recommends a maintenance action (Inspect / Lubricate / Replace Part / Continue Operation) across simulated machine episodes. Show the Q-table after training, the learned policy, and integrate it with the classifier output into a single end-to-end decision pipeline (risk prediction -> action recommendation).

Question 1 - Decision Tree vs Neural Network Classifier

Aim

To implement and compare a **Decision Tree classifier** and a **Neural Network (Multi-Layer Perceptron)** classifier for predicting equipment failure using the same manufacturing equipment dataset.

Dataset

The dataset contains sensor and operational information collected from manufacturing equipment. The main input features are:

- **Vibration**
- **Temperature**
- **Operating Hours**
- **Load Percentage**

The target variable is **Equipment Failure**, which indicates whether the machine is likely to fail.

Methodology

Two classification models are trained using the same training and testing data.

1. Decision Tree Classifier

The Decision Tree classifies equipment based on a sequence of conditions on the input parameters. For example, it may first check whether the **temperature is greater than a particular value**, followed by vibration or load conditions.

The tree is trained using the equipment sensor data, and its performance is evaluated on the test data. The first **three levels of the Decision Tree** are plotted to visualize how the model makes decisions.

2. Neural Network Classifier

A **Multi-Layer Perceptron (MLP)** is used as the Neural Network classifier. The sensor parameters are given as input to the network. The network contains an input layer, one or more hidden layers, and an output layer.

The Neural Network learns complex relationships between equipment sensor values and failure conditions. After training, it predicts whether the equipment will fail using the test dataset.

Performance Metrics

Both classifiers are compared using the following evaluation metrics:

- **Accuracy:** Measures the overall percentage of correct predictions.
- **Precision:** Measures how many predicted failures were actually failures.
- **Recall:** Measures how many actual failures were correctly detected.
- **F1-Score:** Provides a balance between precision and recall.

Visualization

The experiment includes two main visualizations:

Decision Tree: The first three levels of the trained Decision Tree are plotted to show the feature-based decision-making process.

Neural Network: The architecture of the Multi-Layer Perceptron is represented using its input, hidden, and output layers.

Result

The Decision Tree and Neural Network classifiers are successfully implemented using the same equipment dataset. Their performance is compared using Accuracy, Precision, Recall, and F1-Score. The model with the better evaluation metrics can be selected as the preferred classifier for the equipment failure prediction stage of the integrated system.

Conclusion

The experiment demonstrates two different approaches for equipment failure classification. The Decision Tree provides an easily understandable decision-making structure, while the Neural Network can learn more complex relationships between sensor parameters. Comparing their performance helps identify the more suitable model for predicting equipment failures in the manufacturing environment.

Question 2 - Reinforcement Learning for Maintenance Scheduling

Aim

To implement a Q-Learning agent that recommends a suitable maintenance action based on the failure-risk output obtained from Experiment 1. The available actions are Inspect, Lubricate, Replace Part, and Continue Operation.

Methodology

The failure prediction model from Experiment 1 produces the machine's failure-risk level. This risk is used as the initial state for the Q-Learning agent.

The possible states are:

- Low Risk
- Medium Risk
- High Risk

The available maintenance actions are:

- Inspect
- Lubricate
- Replace Part
- Continue Operation

The Q-Learning agent interacts with simulated machine episodes and learns the best action for each risk state. Rewards are assigned according to the suitability of the selected maintenance action.

The Q-value is updated using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where α is the learning rate, γ is the discount factor, r is the reward, s is the current state, and a is the selected action.

Q-Table After Training

After training, the Q-table contains the learned values for each risk state and maintenance action.

Risk State	Inspect	Lubricate	Replace Part	Continue Operation
Low Risk	___	___	___	___
Medium Risk	___	___	___	___
High Risk	___	___	___	___

The action with the highest Q-value in each row is selected as the recommended action.

Learned Policy

Risk State	Recommended Action
Low Risk	Continue Operation
Medium Risk	Inspect / Lubricate
High Risk	Replace Part

The exact policy depends on the rewards and training performed by the Q-Learning agent.

End-to-End Decision Pipeline

Sensor Data



Decision Tree / Neural Network Classifier



Failure-Risk Prediction



Low / Medium / High Risk

↓
Q-Learning Agent
↓
Maintenance Action

For example:

High vibration + High temperature + High load
→ High Failure Risk
→ Q-Learning Agent
→ Replace Part

Normal sensor values
→ Low Failure Risk
→ Q-Learning Agent
→ Continue Operation

Result

The Q-Learning agent learns a maintenance policy through simulated machine episodes. The trained Q-table shows the preferred action for each failure-risk state. The classifier and Q-Learning agent are integrated to create an end-to-end system that converts sensor readings into a failure-risk prediction and then recommends an appropriate maintenance action.

Conclusion

The experiment demonstrates the integration of machine learning and reinforcement learning for predictive maintenance. The classifier predicts the equipment failure risk, while the Q-Learning agent uses this risk to select a suitable maintenance action. This approach can help reduce unexpected failures, maintenance costs, and machine downtime.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Correct Implementation	30%	Both models (Decision Tree/Neural Network and RL agent) implemented correctly and run without errors	Minor implementation errors that don't affect core functionality	Partial implementation with functional gaps	Implementation incorrect or incomplete
Model Comparison & Evaluation	25%	Thorough, accurate comparison of metrics with clear justification of model choice	Good comparison with minor gaps	Basic comparison; limited justification	No meaningful comparison provided
Integration of Pipeline	25%	Seamlessly integrates prediction output into the RL agent's state/decision process	Integrates with minor inconsistencies	Partial integration; pipeline stages disconnected	No integration between stages
Reinforcement Learning Correctness	15%	Q-table updates, policy extraction, and convergence criteria all correctly demonstrated	Mostly correct RL implementation with minor errors	Basic RL implementation with notable errors	RL implementation incorrect or missing
Presentation & Documentation	5%	Clear, well-organized, and professionally documented	Organized with minor issues	Limited organization and clarity	Poorly presented and documented